INCH-POUND

MIL-P-48698 (AR)
AMENDMENT 3
3 May 1989
SUPERSEDING
AMENDMENT 2
28 November 1988

MILITARY SPECIFICATION

PROBE AND S&A ASSEMBLY, TOW-2/E6

This Amendment forms a part of Military Specification MIL-P-48698 (AR), dated 27 April 1987, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1 Between MIL-STD-105 and DOD-STD-2000, add the following:

"MIL-STD-1169 Packaging, Packing and Marking for Shipment of Inert Ammunition Components"

PAGE 3

3.6, Line 3: Delete "is."

PAGE 4

3.18, Lines 2 and 6: Delete "0.5 ohms" and substitute "1.0 ohms."

PAGE 5

* 3.23, Workmanship: Delete paragraph in its entirety and substitute the following:

"3.23 Workmanship.

3.23.1 Metal parts. The TOW 2/E6 Probe and S&A Assembly and its components shall be fabricated and finished in a thorough, workmanlike manner and shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. Wires shall not be damaged during or after assembly. Solder connections shall be in accordance with paragraph 3.20. The cleaning methods used shall not be injurious to any part, nor shall parts be contaminated by the cleaning agents. All required marking shall be clearly legible."

AMSC N/A

FSC 1336

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3.23.2 Plastic parts. Parts shall be free from blisters, cracks, foreign inclusions, scratches, voids or any other surface or internal defect which may detrimentally affect serviceability or appearance, and shall be uniform in color and appearance."

* 3.24, Add new paragraph as follows:

3.24 Process documentation. Concurrent with submission of the First Article Sample (See paragraph 4.3), the contractor shall submit a written description of the manufacturing process used for the vydax lubrication of the wedge. The manufacturing process shall be reviewed by and must be acceptable to the government. The contractor's written plan of process and process controls shall be prepared in accordance with Data Item description DI-P-1604 (Tailored) (see 6.9). Proposed changes to the process which produced the accepted First Article Sample shall be submitted to the government prior to actual implementation of the change. A new First Article Sample may be required for each proposed process change as determined by the Government."

PAGE 6

* 4.3.1, Add at end of paragraph as follows:

"The first article sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packing shall be the same as used in regular production and shall be obtained from the same source of supply. A First Article Sample of portion thereof, as directed by the procuring agency shall also be submitted when ever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity or quality as determined by the Contracting Officer."

4.3.2, Line 4: Delete "byu" and substitute "by."

PAGE 41

4.4.2.31, Major 114: Delete paragraph in its entirety.

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4.4.2.31, Major 115: Change paragraph to the following, "Height of tab straight shear, (.015 min.), (See Note 1)"

PAGE 42

4.4.2.31, Minor 208: Delete "3.2" and substitute "3.23."

PAGE 46

4.4.2.33, Major 124: Delete paragraph in its entirety.

4.4.2.33, Major 125: Change paragraph to the following: "Height of tab straight shear, (.015 min), (See Note 1)"

PAGE 50

4.4.2.35, Minor 211: Delete "3.2" and substitute "3.23."

PAGE 53

4.4.2.37, Minor 206: Delete "3.2" and substitute "3.23."

PAGE 76 & 77

4.4.2.60: Next Higher Assembly: Delete "9398046" and substitute "9388046."

PAGE 77a

4.4.2.61, Add new paragraph as included.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.61	Mechanism Assembly	1 of 1	9388005
			NEXT HIGHER ASSEMBLY: 9387141

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Leaf spring in proper position		100%	3.2	Visual
102	Presence of shoulder washers		100%	3.2	Visual
103	Presence of piston actuator spacer		100%	3.2	Visual
MINOR					
201	Evidence of poor workmanship		1.00%	3.23	Visual

NOTES:

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PAGE 79

* Add the following note to the Lot Quality Confirmation Test:

"If only one defect occurs during the Breakaway Strength Test, the same defective unit shall be retested by the contractor for the Breakaway Strength Requirement. Lot acceptance can be granted only when the retest unit meets in full compliance with the test requirements. When retest unit fails to meet the test requirement."

a. Physical Configuration Audit (PCA) on the defective unit shall be performed by the contractor to ensure configuration.

b. Failure analysis report on the defective unit shall be submitted to the government for review and approval prior to the disposition of the lot."

PAGE 82

4.5.2, After the last sentence add: "(See 6.8)."

4.5.3, After the last sentence add: "(See 6.8)."

PAGE 84

* 4.5.7, Add to end of paragraph the following sentence:

"A record of actual deployment time for each probe shall be maintained for the lot quality confirmation testing (LQCT)."

PAGE 90

4.5.14.2.a, After the first sentence add the following:

"This test shall be conducted prior to continuity circuit tests described in 4.5.15 (Table V). The test circuit shall not exceed 40 microamperes."

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TABLE IV. Delete in its entirety and substitute the following:

"TABLE IV - Open circuit tests

TEST	Resistance ≥ 1.0 megohm FROM (+)	TO (-)
1. Detonator switch*	red cable	inner tip
2. Tip switch	inner tip	black cable
3. Ogive switch	base contact	ogive (or rivet)

*WARNING: If test 1 fails STOP immediately. The S&A is unsafe and should be removed from the lot and disposed of."

4.4.15.2.a, Line 2: Delete "40 microamperes" and substitute "2 milliamperes."

PAGE 91

Table V, Column 3: Delete "0.5 ohms" and substitute "1.0 ohms."

5. PACKAGING., Delete in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation and packaging. The probe and its associated components shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing. Packing of the probe and its associated components shall be in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. The probe and its associated components shall be shipped in the stowed condition in accordance with notes 16 and 17 of Dwg. 9387141."

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* 6.4, Add the following:

"3.24 Manufacturing Methods Report DI-P-1604"

PAGE 93

6.8 Add new paragraph as follows:

"6.8 Tip assembly. Upon insertion of the inert charge assembly, inner tip assembly, and crush ring, apply tip assembly to the probe with 30 ±1 in-lb torque and cure at ambient for 24 hours before commencing first article or lot quality confirmation testing."

* 6.9 Add new paragraph as follows:

6.9 Process documentation (see 3.24). One copy of the plan shall be prepared and forwarded to the following recipients: Commander, U.S. Army, ARDEC, ATTN: SMCAR-CCH-A, Picatinny Arsenal, NJ 07806-5000 and Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAT-B (D), Picatinny Arsenal, NJ 07806-5000.

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